XRPL Validator — Daily Engineering Report · 2026-06-16

Period: 2026-06-16 00:00 to 24:00 US Eastern (2026-06-16 04:00 to 2026-06-17 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 6 commits (0 with a recorded merge into main; xrpl-validator merge records begin 2026-07-04).

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-06-16): 0.85924 (3,036 missed of 21,568). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merge records begin 2026-07-04; this day's commits are listed by commit date below.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(ffi): 3.2.0 header paths for HashRouter/LoadFeeTrack/OrderBookDB** (`2f0f58c`, xrpl-validator) 3.2.0 relocated these out of xrpld/app/*: HashRouter->xrpl/core, LoadFeeTrack->xrpl/server, OrderBookDB->xrpl/ledger.
- **fix(ffi): NoopOrderBookDB for 3.2.0 (OrderBookDB now pure-virtual)** (`b9cf3ce`, xrpl-validator) 3.2.0 turned OrderBookDB into a 5-method interface; the concrete OrderBookDBImpl needs a full ServiceRegistry. Single-tx apply crosses offers via the ledger View's book-directory SLEs, not this cache, so a no-op impl is hash-safe.
- **fix(ffi): default HashRouter::Setup (setup_HashRouter moved to xrpld in 3.2.0)** (`13e6ecc`, xrpl-validator)
- **feat(lockstep): env-gated ledger_hash shadow eval at the FFI hook (M1 task 4)** (`3ff0144`, xrpl-validator) check_shadow_hash now returns the computed state root (Option<[u8;32]>) instead of bool; the ws_sync FFI-shadow hook feeds it + header fields into lockstep::evaluate_ledger when XRPL_LOCKSTEP_SHADOW=1, recording a process-wide LockstepMeter. Shadow-only, no signing/state impact.
- **feat(lockstep): expose xrpl_lockstep_shadow_match_ratio gauge (M1 task 5)** (`294e85c`, xrpl-validator)
- **fix(peer): raise codec MAX_MESSAGE_SIZE 4MiB -> 64MiB to match rippled 3.2.0** (`7e5b1b9`, xrpl-validator) 3.2.0 peers legitimately send bulk frames up to kMaximumMessageSize=megabytes(64) (overlay/Message.h; 26-bit payload header field). The stale 4 MiB cap silently dropped ~60 MB messages from many peers (116 codec errors observed post-upgrade). Not consensus-critical (state syncs from the reference node, not peer bulk) but correct protocol behavior. Tests: oversized boundary now 65 MiB; new test asserts sub-cap accept.

Independence program

No independence record for UTC day 2026-06-16; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
06-16 04:03	—	xrpl-validator	`2f0f58c`	fix(ffi): 3.2.0 header paths for HashRouter/LoadFeeTrack/OrderBookDB
06-16 04:05	—	xrpl-validator	`b9cf3ce`	fix(ffi): NoopOrderBookDB for 3.2.0 (OrderBookDB now pure-virtual)
06-16 04:06	—	xrpl-validator	`13e6ecc`	fix(ffi): default HashRouter::Setup (setup_HashRouter moved to xrpld in 3.2.0)
06-16 16:13	—	xrpl-validator	`3ff0144`	feat(lockstep): env-gated ledger_hash shadow eval at the FFI hook (M1 task 4)
06-16 16:13	—	xrpl-validator	`294e85c`	feat(lockstep): expose xrpl_lockstep_shadow_match_ratio gauge (M1 task 5)
06-16 16:27	—	xrpl-validator	`7e5b1b9`	fix(peer): raise codec MAX_MESSAGE_SIZE 4MiB -> 64MiB to match rippled 3.2.0

Times in UTC. A commit appears on the day it was committed to main or merged into it.